Antony Manuel

Paris 75011, France | +33 (0) 7 43 67 74 16 | amanuel.jn@gmail.com | |

Education

IMT Nord Europe | Master of Machine Learning — Lille
Hilbert Spaces, Graph Theory, Statistics, Optmization, NLP, Computer Vision — *Sep. 2023 – Sep. 2026*

University of Paris Cité | Double Bachelor of Physics and Chemistry — Paris
Quantum Physics, Electromagnetism, Atomistic, Analytical & Organic Chemistry — *Sep. 2020 – Sep. 2022*

Experience

Research Scientist Intern — May – Aug. 2025
Numalis — *Montpellier, FR*

- Conducted research in the R&D team on adversarial robustness of deep learning models
- Developed a detection framework based on explainability to identify adversarial examples and improve model interpretability
- Designed and implemented experimental prototypes, achieving 97% detection accuracy against FGSM attacks
- Analyzed the interaction between interpretability techniques and model resilience, providing insights into robust AI system design
- Authored research documentation and presented findings internally and at technical meetings
- Contributed to the development of methods for integrating explainable defenses into production ML pipelines

Machine Learning Engineer Intern — May – Aug. 2024
Veolia — *Paris, FR*

- Engineered a Streamlit-based user interface for Veolia's intranet chatbot to facilitate seamless employee interaction and query handling
- Developed a multimodal data processing pipeline to extract and structure information from internal *PDF*, image, and video documents
- Implemented a RAG architecture to enhance LLM response accuracy and contextual relevance
- Integrated FAISS vector indexing with Sentence Transformers for strong performance semantic search and source attribution
- Optimized end-to-end retrieval and generation pipeline, improving chatbot response accuracy from 34% to 71%

Projects

Quantum Generative Adversarial Network | *JAX, PyTorch, Pennylane* — Nov. 2025 – March 2026

- Conducted independent research on generative modeling using classical and quantum adversarial learning frameworks
- Developed and evaluated classical, quantum, and hybrid GANs to study expressivity and learning behavior
- Analyzed the impact of parameterized quantum circuits on adversarial training stability and generative performance

Explainability of Musical Harmony, INRIA Lab | *Python, PyTorch, HuggingFace* — Sep. 2025 – Feb. 2026

- Implemented Transformer-based architectures and applied NLP techniques to model rhythmic patterns and temporal dependencies in musical sequences
- Applied LRP to interpret model attention and feature attribution, revealing how neural networks learn rhythmic structure

Enzyme and Sign language Recognition | *Python, PyTorch* — Oct. – Dec. 2025

- Developed and optimized Graph Convolutional Networks for molecular and spatial skeleton graph classification
- Engineered data preprocessing pipelines to transform molecular data and image coordinate into optimized graph representations

Skills

Languages: Fluent English, Intermediate Spanish, Intermediate Mandarin
Programming Languages: Python, SQL, Java, JavaScript, HTML/CSS
Developer Tools: Git, GCP, AWS, VS Code
Libraries: PyTorch, Tensorflow, Keras, JAX, Qiskit, PennyLane